Disaster Data Engineering & Intelligence Platform

Real USGS earthquake and NOAA storm event data — validated end-to-end pipeline

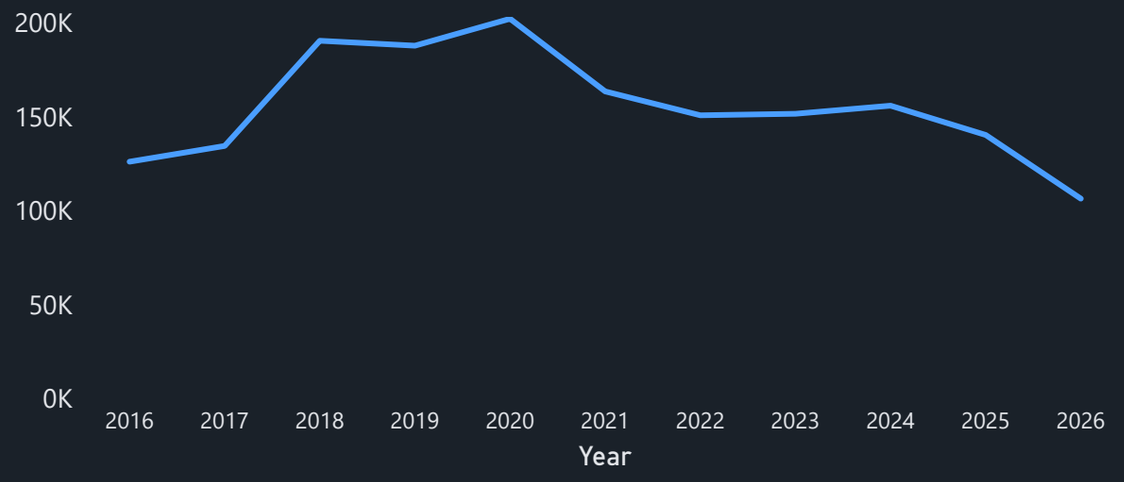
Overview | Earthquakes | Storms | Platform

1.7M	693.0K	10.4K	20.1K	$248.80B
EARTHQUAKES ANALYZED	STORMS ANALYZED	STORM DEATHS	STORM INJURIES	STORM PROPERTY DAMAGE

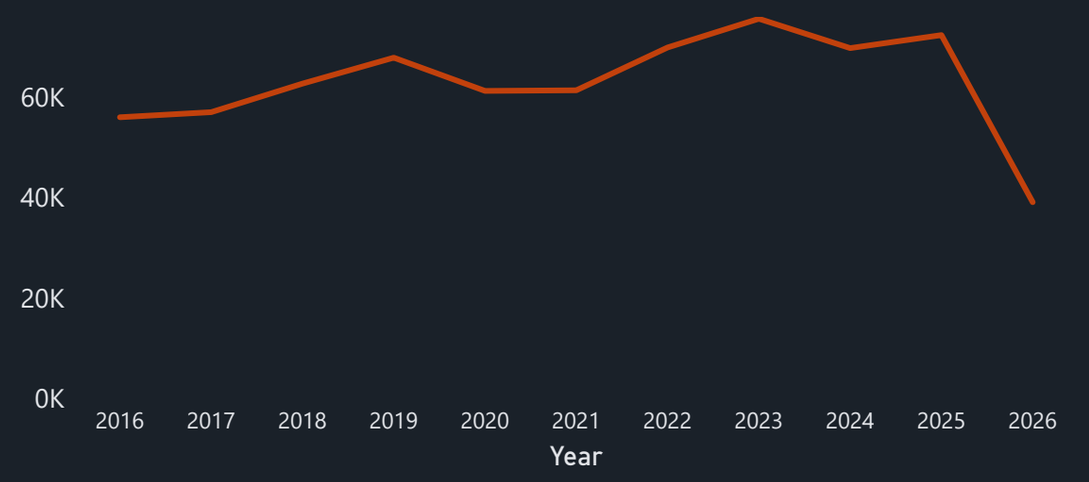
EARTHQUAKE DATA: Jan 2016 to Oct 2026

STORM DATA: Jan 2016 to Jul 2026

Earthquake Activity Over Time



Storm Activity Over Time



Data through 1 Oct 2026 (earthquakes) and 1 Jul 2026 (storms). The latest year is year-to-date and updates with each pipeline run.

Note: earthquakes and storm events are analyzed independently. No direct event-level correlation is drawn between the two sources, as no defensible geographic or temporal join key exists between USGS and NOAA records.

Earthquake Analysis

Magnitude, depth and geographic characteristics of 1.7M real earthquakes, 2016 to 2026

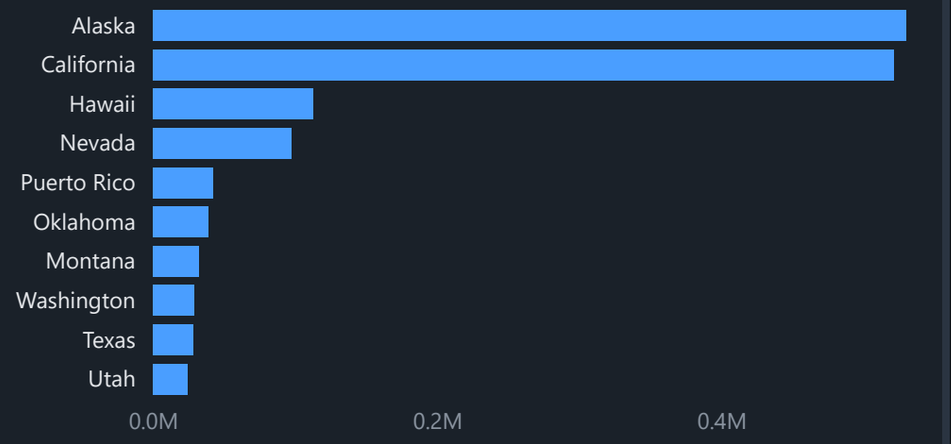
1,434
SIGNIFICANT EARTHQUAKES (M ≥ 6.0)

0.08%
% OF ALL EVENTS

1.63
AVERAGE MAGNITUDE

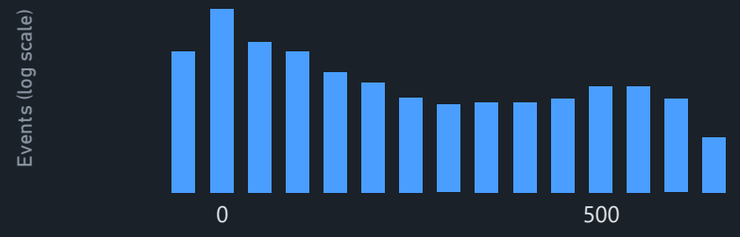
8.80
STRONGEST EVENT (MAGNITUDE)

Top 10 Regions by Earthquake Count



Year: All

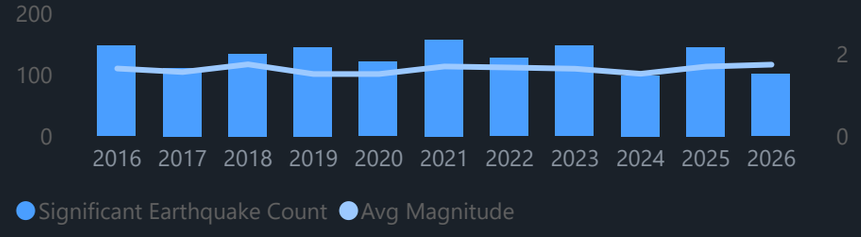
Depth Distribution (km)



Event Locations



M6+ Events & Average Magnitude by Year



Where	Strongest	Peak Year
Alaska leads with 31% of all recorded events	Strongest event: M8.8, 2025 Kamchatka Peninsula, Russia Earthquake (2025)	Peak M6+ year: 2021 with 157 significant earthquakes

Storm Analysis

Event types, seasonality and impact across 693K real storm events, 2016 to 2026

Overview | Earthquakes | **Storms** | Platform

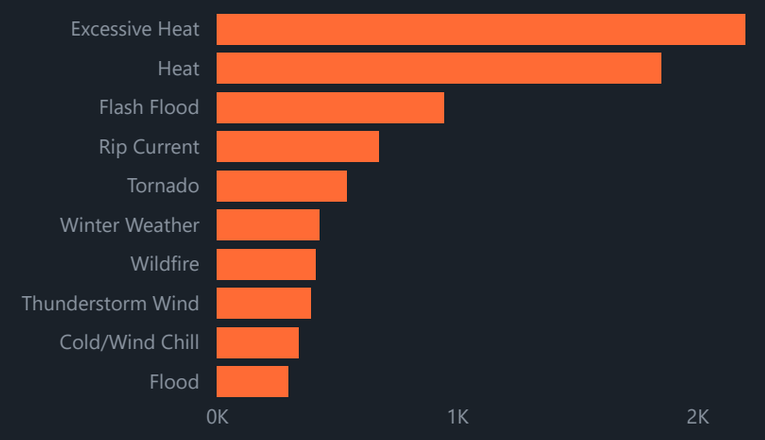
10.4K
TOTAL DEATHS

20.1K
TOTAL INJURIES

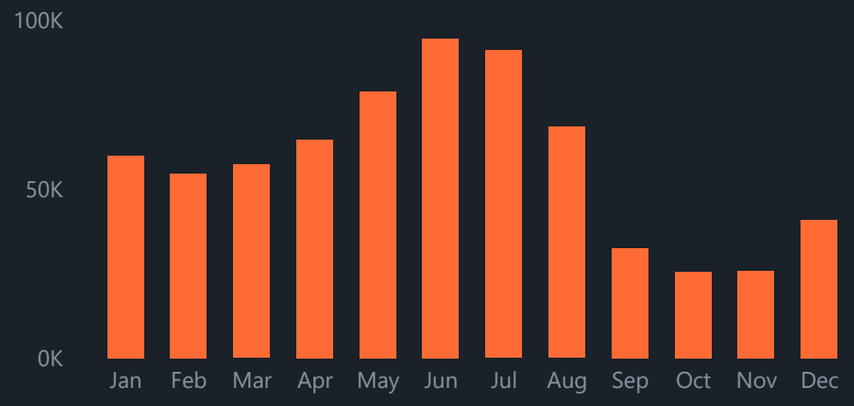
$248.80B
PROPERTY DAMAGE ($)

Thunderstorm Wind
MOST FREQUENT EVENT TYPE

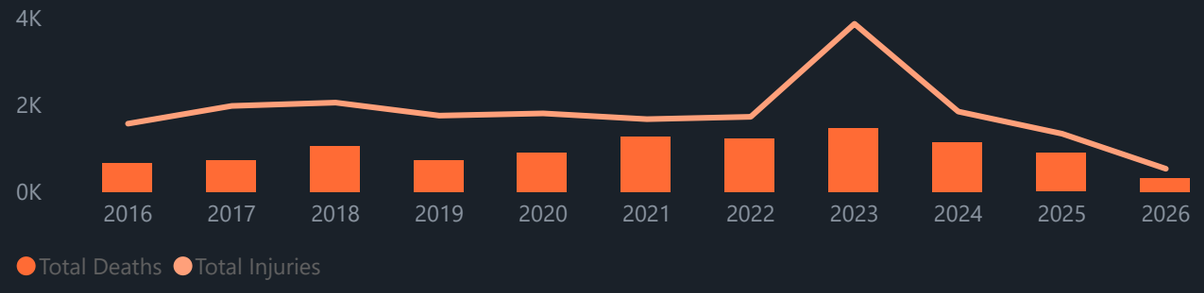
Top 10 Deadliest Event Types



Storm Events by Month



Deaths & Injuries by Year



Event Type
All

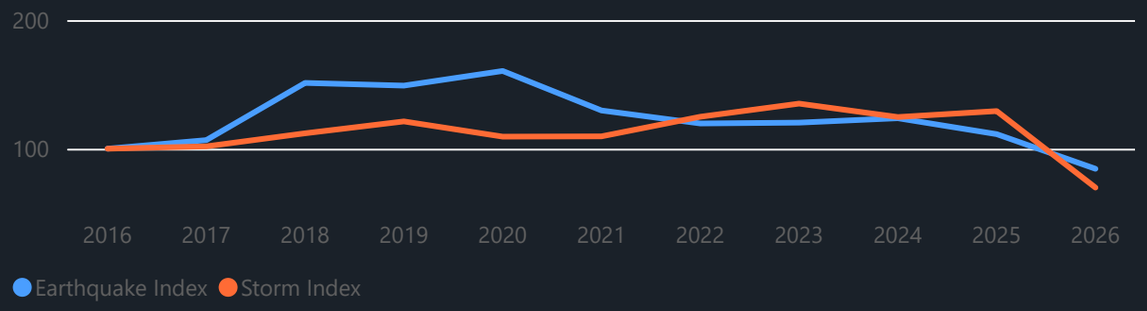
Month
All

Year
All

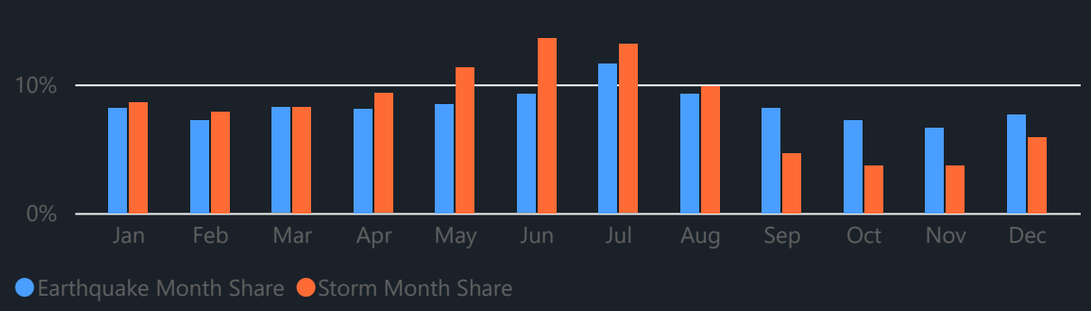
Deadliest	Injuries	Peak Season
Excessive Heat is the deadliest event type, with 2,199 deaths	Injuries peaked in 2023 with 3,857 recorded	Storm activity peaks in Jun (94,404 events across the period)

Platform Overview

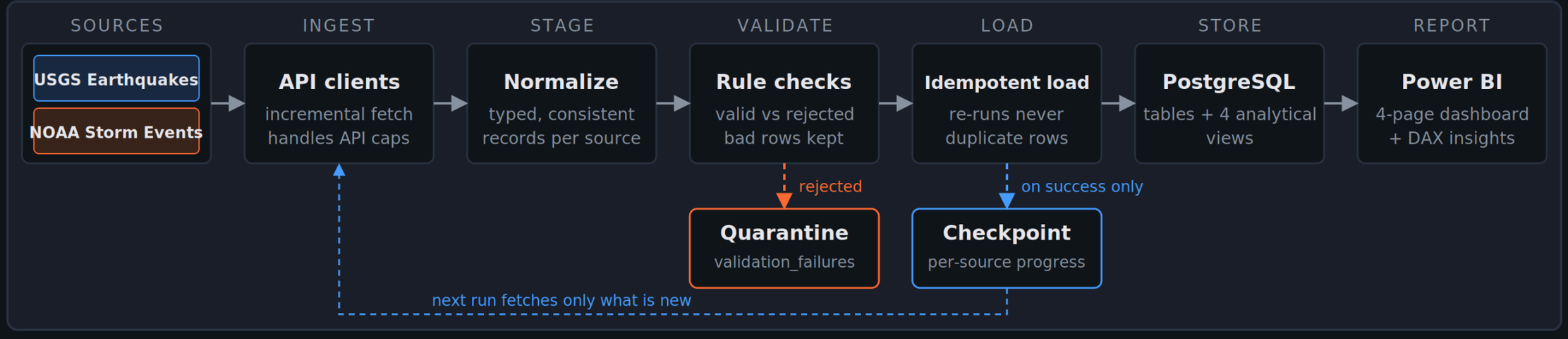
Activity vs 2016 (2016 = 100)



Seasonality: Share of Events by Month



DATA PIPELINE



Earthquakes	Storms	Seasonality
Earthquake activity peaked in 2020 (+60% vs 2016)	Storm activity peaked in 2023 (+35% vs 2016)	Storms peak in Jun (14% of events); earthquakes stay between 7% and 12% every month